

Demystifying ML Evaluation

A Beginner's Guide to Understanding Accuracy, Precision, Recall, F1-Score, and Feature
Scaling

Why Do We Need to Evaluate Models?

Building a model isn't enough. Evaluation tells us how well our model performs on new, unseen data. It helps us choose the best model and, more importantly, trust our model to make correct decisions.

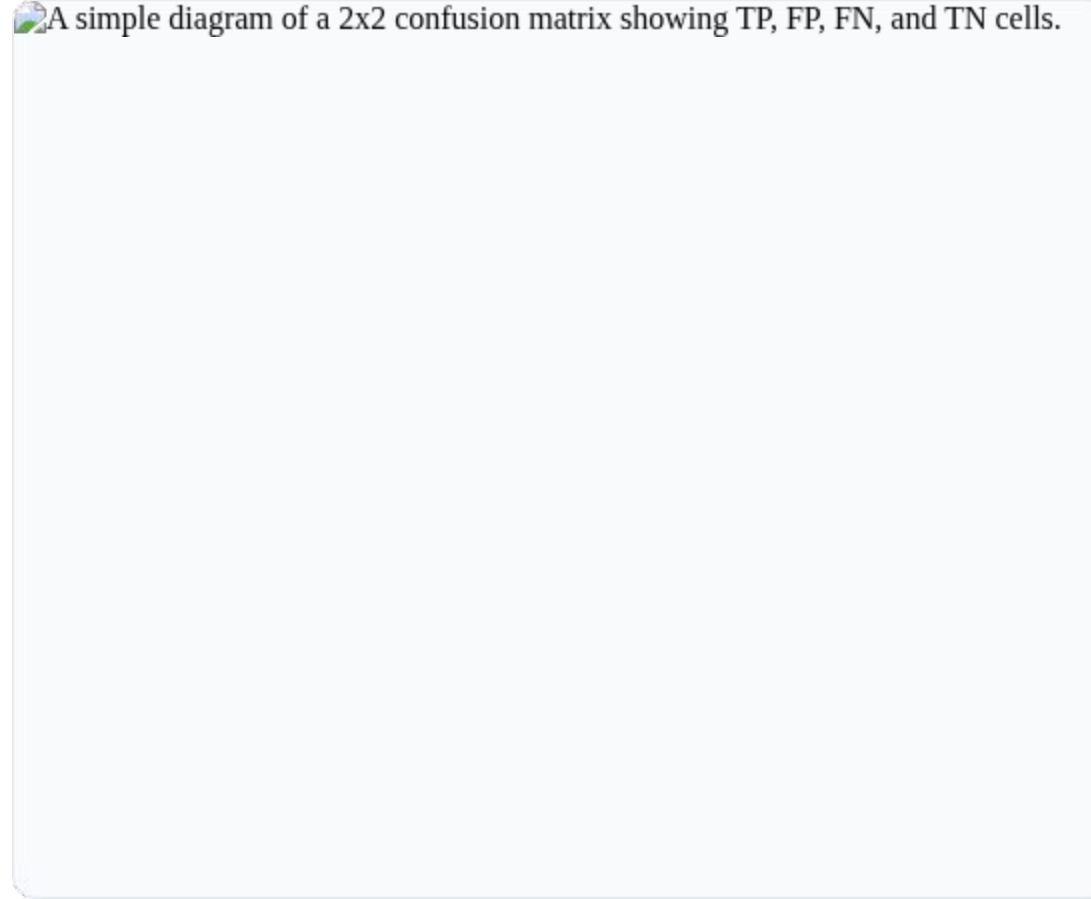
The Confusion Matrix: Our Scorecard

Before we get to metrics, we need the **Confusion Matrix**.

It's a simple table that compares our model's predictions to the actual truth.

It's the foundation for almost all classification metrics.

- **True Positive (TP):** Correctly predicted as positive.
- **True Negative (TN):** Correctly predicted as negative.
- **False Positive (FP):** Wrongly predicted as positive.
- **False Negative (FN):** Wrongly predicted as negative.

A simple diagram of a 2x2 confusion matrix showing TP, FP, FN, and TN cells.

Metric 1: Accuracy



Definition

The percentage of all predictions that the model got right. The most intuitive metric.



Formula

All "Correct" / All Predictions

$$\frac{TP + TN}{TP + TN + FP + FN}$$



Pros & Cons

Pro: Simple to understand and explain.

Con: Very misleading on "imbalanced" datasets.

The Problem with Accuracy

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Imagine a dataset with 100 people: 99 are healthy, 1 has a rare disease. A "lazy" model that predicts 'Healthy' every time is 99% accurate... but 100% useless for finding the disease.

— This is the "Accuracy Paradox." This is why we need other metrics.

Metric 2: Precision

The Question

"Of all the times we predicted 'Positive', what percentage was *actually* positive?"

Precision is about being **correct** when you *do* predict positive.

Formula

$$\frac{TP}{TP + FP}$$

Use Case: High Precision

Critical when the cost of a **False Positive (FP)** is high.

Example: Email Spam Filter

You would rather let one spam email (FN) into your inbox than have one important email (a False Positive) sent to spam. High precision ensures that what it *does* call spam is *very likely* to be spam.

Metric 3: Recall

The Question

"Of all the ***actual*** positive cases, what percentage did we ***find*** (or 'recall')?"

Recall is about **finding** as many of the positive cases as possible.

Formula

$$\frac{TP}{TP + FN}$$

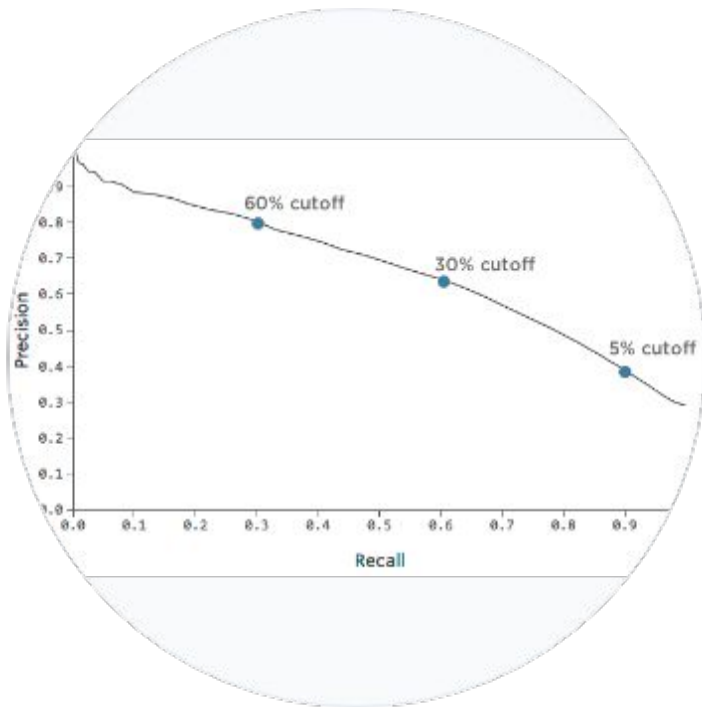
Use Case: High Recall

Critical when the cost of a **False Negative (FN)** is high.

Example: Medical Diagnosis

You ***cannot*** miss a real disease (a False Negative). It's better to have a few false alarms (FP, low precision) that can be cleared by another test, than to miss a sick patient. High recall ensures you find most of the real cases.

The Precision-Recall Trade-off



You Can't Always Have Both

Increasing precision often means being more "careful," which can cause you to miss some cases (lowering recall).

Increasing recall often means trying to find "everything," which can lead to more false alarms (lowering precision).

The balance you choose depends on your problem: Do you hate false positives (need precision) or false negatives (need recall) more?

Metric 4: F1 Score

The Best of Both Worlds

The **F1 Score** is the **Harmonic Mean** of Precision and Recall.

It's a single number that balances both metrics. It's the best metric to use if you have an imbalanced dataset (like our disease example) and you care about both False Positives and False Negatives.

Formula

A high F1 score means you have a good balance of both precision and recall.

$$2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

Shifting Gears: Evaluating Regression

So far, we've covered **Classification** (predicting a category like 'Spam' or 'Healthy'). What if we want to predict a *number*, like a house price?

Metric 5: Mean Absolute Error (MAE)



What Is It?

Used for **Regression**. It's the average *absolute* difference between the model's prediction and the actual value. It answers: "On average, how far off is my prediction?"

Example

Actual House Price: \$300k.
Model Predicts: \$310k.
The **Absolute Error** is \$10k.
MAE is the average of these errors across your whole dataset.



Interpretation

Easy to understand because it's in the same units as your target.
An **MAE of \$5k** means your price predictions are, on average, \$5k off from the real price.

A Different Topic: Feature Scaling



What Is It?

A **pre-processing step**, not an evaluation metric. It's the process of adjusting the range of your data features (e.g., Age, Salary) to be on a similar scale.



Why Do It?

Many models are "distance-based" (like k-NN, SVM). If one feature (Salary: 50,000) is 1,000x larger than another (Age: 50), the model will be biased and think Salary is 1,000x more important.

Scaling fixes this.



Who Needs It?

YES: k-NN, SVM, K-Means, Neural Networks, Gradient Descent.

NO: Tree-based models (like Decision Trees, Random Forest) are not affected by scale.

Feature Scaling Example (Min-Max Scaling)

We scale all values to be in a common range, like [0, 1]. Notice how "Salary" no longer dominates "Age" in the "After" columns.

Feature	Original Value 1	Original Value 2	Scaled Value 1	Scaled Value 2
Age	25	60	0.25	0.60
Salary	50,000	200,000	0.10	0.40
(Note)	(Scales are very different)	(Scales are comparable [0-1])		

Questions?

Thank you for your attention.

Image Sources



<https://i.sstatic.net/BlmMF.png>

Source: stackoverflow.com



<https://www.qualtrics.com/m/assets/support/wp-content/uploads/2017/07/precision-recall-curve-for-regression-docs1.png>

Source: www.qualtrics.com